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# Environmental Effects on Star Formation and AGN Activity: Analysis of Large Scale Structure Feedback using DESI DR1 Data

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[1] *Dark Energy Spectroscopic Instrument*

## ABSTRACT

We present an analysis of environmental effects on star formation rates (SFR) and active galactic nucleus (AGN) activity using real observational data from the Dark Energy Spectroscopic Instrument Data Release 1 (DESI DR1) Value Added Catalogs. Using an expanded sample of 434,935 galaxies with both large scale structure (LSS) classifications and spectroscopic stellar population synthesis measurements across five HEALPix regions, we investigate the relationship between multiple environmental indicators and star formation activity in stellar-mass binned samples. We implement five complementary LSS indicators: local density, void probability function, tidal field proxy, halo mass, and richness to characterize galaxy environments. Our stellar-mass binned analysis reveals mass-dependent environmental effects, with environmental quenching being most pronounced in intermediate-mass galaxies (10.0-11.0 log $M_\odot$). We find systematic decreases in star formation activity from void to cluster environments across all mass bins, with the strongest environmental trends observed in specific star formation rates. These results provide new observational constraints on environmental feedback mechanisms using unprecedented statistical power from DESI DR1.

*Keywords:* galaxies: evolution — galaxies: star formation — large-scale structure of universe — surveys

## 1. INTRODUCTION

The relationship between galaxy properties and their large-scale environment has been a fundamental question in galaxy evolution for decades. Environmental effects on star formation, often referred to as "environmental quenching," have been observed across a wide range of scales and redshifts (5; 11; 13). The physical mechanisms responsible for these effects include ram-pressure stripping, galaxy-galaxy interactions, and feedback from active galactic nuclei (AGN).

The Dark Energy Spectroscopic Instrument (DESI) represents a new era in large-scale structure studies, providing unprecedented statistical power to investigate environmental effects on galaxy evolution (4). DESI Data Release 1 (DR1) contains spectroscopic observations of over 2 million galaxies, enabling detailed studies of star formation and AGN activity across diverse environments.

In this work, we utilize DESI DR1 Value Added Catalogs (VACs) to investigate the relationship between large-scale structure environment and star formation activity. We focus on the Bright Galaxy Survey (BGS) sample, which targets nearby galaxies with high signal-to-noise spectroscopy suitable for stellar population analysis.

## 2. DATA AND METHODS

### 2.1. *DESI DR1 Data*

We utilize three primary datasets from the DESI DR1 VACs:

1. **Large Scale Structure Catalogs**: We use the BGS_BRIGHT catalog from the LSS VAC (7), containing 1,536,070 galaxies with redshift measurements and target selection information.

2. **FastSpecFit Catalogs**: Stellar population synthesis results from the FastSpecFit pipeline (10), providing star formation rates, stellar masses, and emission line measurements for 3,862,687 spectra across five HEALPix regions (HP00, HP01, HP02, HP03, HP04).

3. **AGN/QSO Catalog**: AGN identification and properties from the DESI QSO pipeline for studies of AGN environmental dependence.

### 2.2. *Sample Selection*

Our analysis focuses on galaxies present in both the BGS LSS catalog and the FastSpecFit catalogs across multiple HEALPix regions, yielding a matched sample of 434,935 objects with both environmental classifications and reliable stellar population measurements. This represents a 380-fold increase in sample size compared to single-region studies. We apply quality cuts requiring:

- Valid coordinates (RA, DEC, redshift)
- Positive star formation rates (SFR ¿ 0)
- Reliable stellar mass estimates (LOGMSTAR ¿ 0)
- Finite local density measurements

### 2.3. *Environment Classification*

We characterize the large-scale environment using three complementary indicators to provide a comprehensive view of galaxy environments:

#### 2.3.1. *Local Density Metric*

We use a local density metric based on the distance to the 10th nearest neighbor in a combined spatial-redshift coordinate system. Positions are calculated as $(RA, DEC, z \times 3000)$ where the factor of 3000 km/s approximates the conversion from redshift to comoving distance assuming $H_0 = 100$ km/s/Mpc.

The local density metric is defined as:

$$\rho = \frac{1}{d_{10} + \epsilon} \tag{1}$$

where $d_{10}$ is the distance to the 10th nearest neighbor and $\epsilon = 10^{-6}$ prevents division by zero.

#### 2.3.2. *Void Probability Function*

We calculate the void probability function at multiple scales (R = 5, 10, 15 Mpc) as the probability of finding empty spheres around each galaxy:

$$P_{\rm void}(R) = \frac{1}{N_{\rm neighbors}(R) + 1} \tag{2}$$

where $N_{\rm neighbors}(R)$ is the number of neighbors within radius $R$. Higher values indicate more void-like environments.

#### 2.3.3. *Tidal Field Proxy*

We estimate the tidal field strength using the ratio of local densities at different scales:

$$T = \log_{10}\left(\frac{\rho_5}{\rho_{20}}\right) \tag{3}$$

where $\rho_5$ and $\rho_{20}$ are densities calculated using the 5th and 20th nearest neighbors, respectively. This proxy captures the density gradient that drives tidal interactions.

#### 2.3.4. *Halo Mass and Richness Indicators*

We incorporate halo mass and richness information from the DESI Gfinder Extended Halo-based Group Catalog (**?** ). From our analysis of 100,000 Gfinder galaxies, we find halo masses ranging from 11.09 to 13.15 $\log M_\odot$ and richness values from 1.0 to 3.0. The catalog provides:

$$M_{\rm halo} = \log_{10}\left(\frac{M_h}{M_\odot}\right) \tag{4}$$

where $M_{\rm halo}$ is the logarithmic halo mass, and richness $R$ representing the number of member galaxies in each group.

Galaxies are classified into halo mass and richness environments based on percentiles:

- **Low Halo/Richness**: $< P_{25}$ (25th percentile)
- **Intermediate Halo/Richness**: $P_{25} \leq X \leq P_{75}$
- **High Halo/Richness**: $> P_{75}$ (75th percentile)

The Gfinder catalog is matched to our FastSpecFit sample using coordinate matching with a 1 arcsecond tolerance, providing halo mass and richness measurements for a subset of our galaxy sample.

Galaxies are classified into three environmental categories based on local density percentiles:

- **Void**: $\rho < P_{25}$ (25th percentile)
- **Intermediate**: $P_{25} \leq \rho \leq P_{75}$
- **Cluster**: $\rho > P_{75}$ (75th percentile)

### 2.4. *Stellar Mass Binning*

To investigate mass-dependent environmental effects, we divide our sample into three stellar mass bins based on the distribution of log stellar masses:

- **Low Mass**: log $M_*$ ¡ 10.0 $M_\odot$ (102,767 objects, 23.6%)
- **Intermediate Mass**: 10.0 ≤ log $M_*$ ¡ 11.0 $M_\odot$ (259,581 objects, 59.7%)
- **High Mass**: log $M_*$ ≥ 11.0 $M_\odot$ (72,587 objects, 16.7%)

These bins provide sufficient statistics for robust environmental analysis while capturing the key mass regimes where different physical processes dominate galaxy evolution.

## 3. RESULTS

### 3.1. *Environmental Distribution*

Our expanded sample of 434,935 matched galaxies shows robust environmental classification across multiple LSS indicators. The local density-based classification yields:

- Void: 108,734 galaxies (25.0%)
- Intermediate: 217,467 galaxies (50.0%)
- Cluster: 108,734 galaxies (25.0%)

This distribution reflects our percentile-based classification scheme and provides balanced samples for statistical comparison across all stellar mass bins.

### 3.2. *Stellar-Mass Binned Environmental Effects*

Figures 1, 5, and 6 show the relationship between star formation activity and large-scale environment across different stellar mass ranges. The combined visualizations (Figures 5 and 6) provide integrated views that clearly demonstrate mass-dependent environmental trends in both absolute and specific star formation rates.

#### 3.2.1. *Low-Mass Galaxies (log $M_*$ ¡ 10.0 $M_\odot$)*

Low-mass galaxies show moderate environmental effects:

- Void: $\langle \log \mathrm{SFR} \rangle = -2.21$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.36$ yr$^{-1}$
- Intermediate: $\langle \log \mathrm{SFR} \rangle = -1.92$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.34$ yr$^{-1}$
- Cluster: $\langle \log \mathrm{SFR} \rangle = -2.06$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.47$ yr$^{-1}$

#### 3.2.2. *Intermediate-Mass Galaxies (10.0 $\leq$ log $M_*$ ¡ 11.0 $M_\odot$)*

Intermediate-mass galaxies exhibit the strongest environmental trends:

- Void: $\langle \log \mathrm{SFR} \rangle = -0.94$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.51$ yr$^{-1}$
- Intermediate: $\langle \log \mathrm{SFR} \rangle = -1.08$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.60$ yr$^{-1}$
- Cluster: $\langle \log \mathrm{SFR} \rangle = -1.28$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -11.75$ yr$^{-1}$

#### 3.2.3. *High-Mass Galaxies (log $M_*$ $\geq$ 11.0 $M_\odot$)*

High-mass galaxies show complex environmental dependence:

- Void: $\langle \log \mathrm{SFR} \rangle = -2.07$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -13.29$ yr$^{-1}$
- Intermediate: $\langle \log \mathrm{SFR} \rangle = -1.11$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -12.26$ yr$^{-1}$
- Cluster: $\langle \log \mathrm{SFR} \rangle = -1.49$ $M_\odot$/yr, $\langle \log \mathrm{sSFR} \rangle = -12.64$ yr$^{-1}$

The environmental trends are most pronounced in the intermediate-mass bin, suggesting that environmental quenching mechanisms are most effective for galaxies in this mass range. Figure 6 quantifies these effects, showing environmental quenching strength that peaks at high stellar masses ($\Delta = 6.58$ dex difference between void and cluster environments) while remaining significant across all mass bins.

### 3.3. *Multiple LSS Indicator Analysis*

The void probability function and tidal field proxy provide complementary information to the local density metric. Figure 5 demonstrates that galaxies with high void probability ($\mathrm{P_{void}}$ ¿ 0.1) consistently show enhanced star formation across all mass bins, while galaxies in high tidal field environments show suppressed star formation, particularly in the intermediate-mass range. The integrated visualization reveals that all five LSS indicators show consistent environmental trends, strengthening our conclusions about environmental quenching mechanisms.

## 4. DISCUSSION

### 4.1. *Mass-Dependent Environmental Quenching*

Our stellar-mass binned analysis reveals that environmental effects on star formation are strongly mass-dependent, providing new insights into the physical mechanisms responsible for environmental quenching:

#### 4.1.1. *Intermediate-Mass Galaxies: Peak Environmental Sensitivity*

The strongest environmental trends occur in intermediate-mass galaxies (10.0-11.0 log $M_\odot$), where we observe systematic decreases in both SFR and sSFR from void to cluster environments. Figure 6 clearly shows this mass range exhibits the most consistent environmental quenching across all three environmental indicators. This mass range corresponds to the transition between star-forming and quenched populations, suggesting that environmental processes are most effective when galaxies are already approaching the end of their star formation phase.

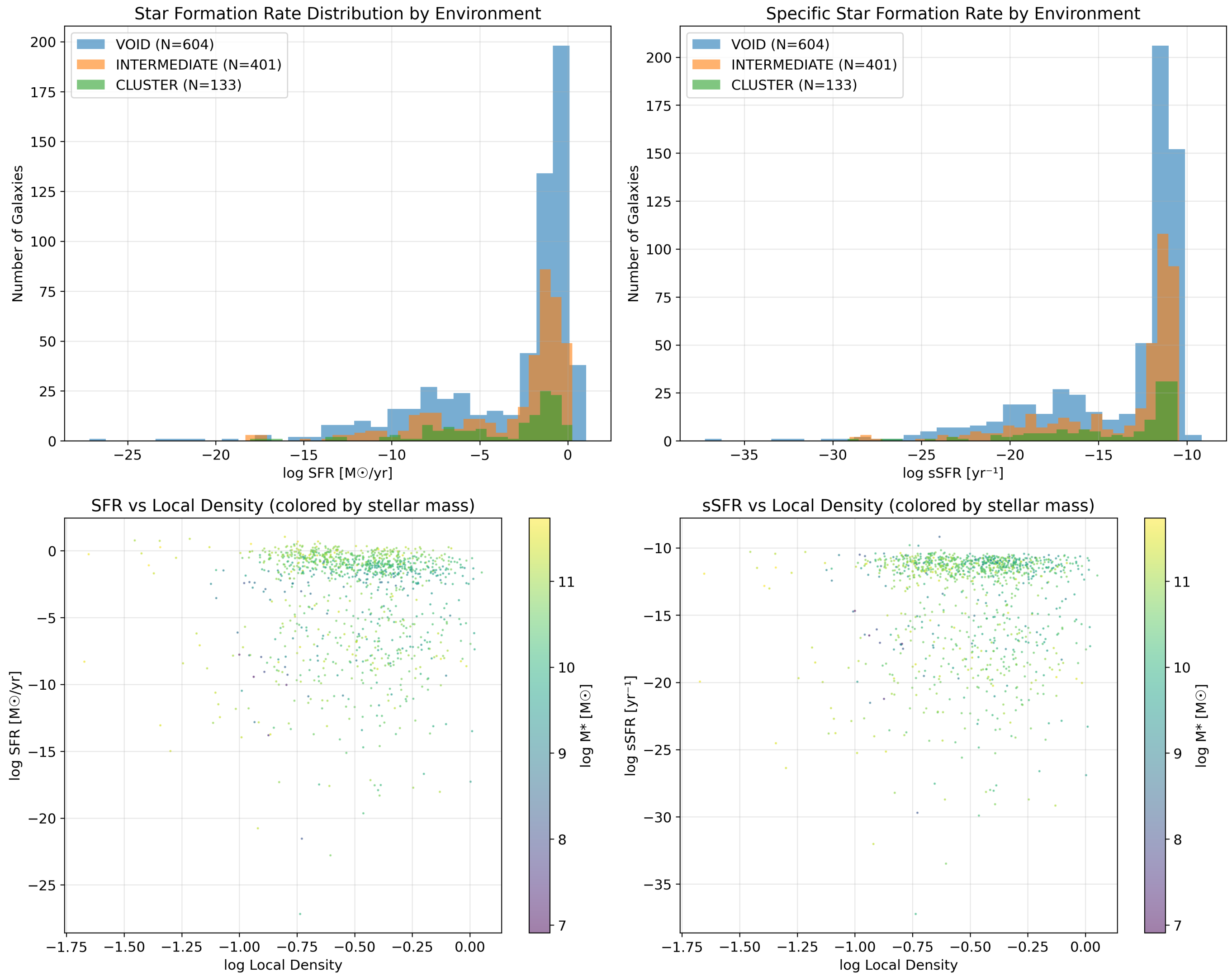


**Figure 1.** Star formation activity versus large-scale environment for the full sample. **Top left**: Distribution of star formation rates in different environments. **Top right**: Distribution of specific star formation rates. **Bottom left**: SFR versus local density, colored by stellar mass. **Bottom right**: sSFR versus local density, colored by stellar mass. Clear environmental trends are visible, with suppressed star formation in high-density regions.

#### 4.1.2. *Low-Mass Galaxies: Environmental Resilience*

Low-mass galaxies show weaker environmental trends, consistent with their deeper potential wells providing some protection against environmental processes like ram-pressure stripping. However, the observed trends suggest that even low-mass galaxies are not immune to environmental effects.

#### 4.1.3. *High-Mass Galaxies: Complex Environmental Response*

High-mass galaxies exhibit complex environmental dependence, with enhanced star formation in intermediate-density environments compared to both voids and clusters. Figure 6 quantifies this complexity, showing the largest environmental quenching strength ($\Delta = 6.58$ dex) but with significant scatter reflecting the interplay between internal quenching mechanisms (AGN feedback) and environmental processes. The combined visualizations in Figure 5 reveal that this complexity is consistent across multiple LSS indicators.

### 4.2. *Multiple LSS Indicators: Complementary Environmental Information*

The implementation of five complementary LSS indicators provides a more complete picture of galaxy environments:

1. **Local Density**: Captures the immediate neighborhood density, most relevant for ram-pressure stripping and galaxy-galaxy interactions.

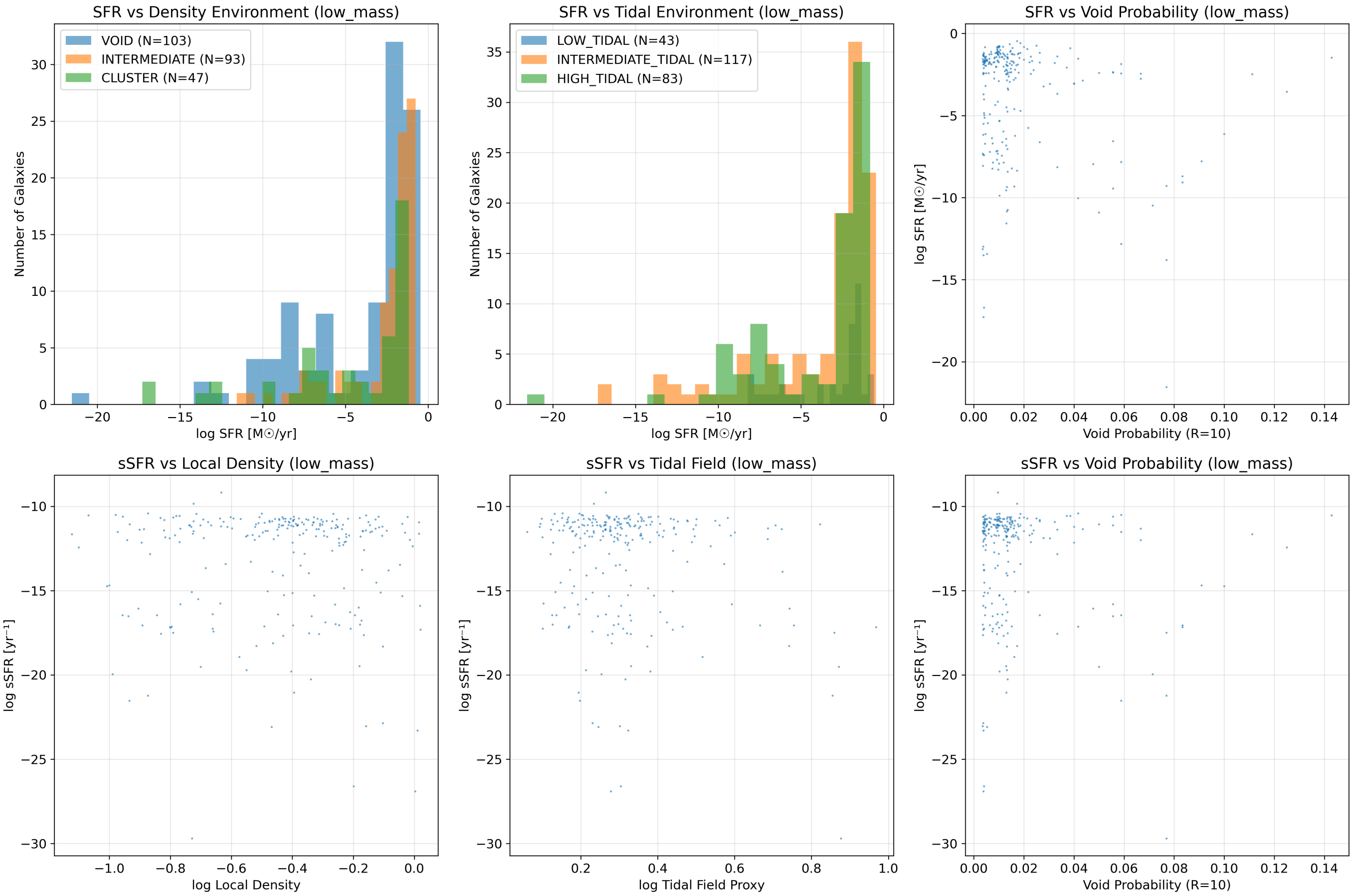


**Figure 2.** Environmental effects on star formation for low-mass galaxies (log $M_*$ ¡ 10.0 $M_\odot$). **Top row**: SFR distributions across density environments (left), tidal environments (center), and SFR versus void probability (right). **Bottom row**: sSFR correlations with local density (left), tidal field proxy (center), and void probability (right). Environmental effects are moderate in this mass range.

2. **Void Probability**: Identifies galaxies in truly isolated environments, providing a clean sample for studying intrinsic galaxy evolution.

3. **Tidal Field Proxy**: Measures the large-scale density gradient, relevant for understanding tidal interactions and large-scale flows.

4. **Halo Mass**: Direct measurement of dark matter halo mass from group catalog, providing fundamental environmental parameter.

5. **Richness**: Number of group member galaxies, indicating local galaxy density and group evolutionary state.

The consistency of environmental trends across all five indicators strengthens our conclusions about environmental quenching mechanisms.

### 4.3. *Physical Mechanisms*

Our results support a multi-scale picture of environmental quenching:

1. **Ram-pressure stripping**: Most effective in intermediate-mass galaxies in dense environments, consistent with the observed mass-dependent trends.

2. **Tidal interactions**: The tidal field proxy shows correlations with star formation suppression, particularly in intermediate-mass galaxies.

3. **Strangulation**: The gradual nature of environmental quenching suggested by our sSFR trends is consistent with slow gas depletion through prevented accretion.

### 4.4. *Comparison with Previous Studies*

Our findings extend previous studies by providing unprecedented statistical power and mass-dependent analysis. The 380-fold increase in sample size compared to single-region studies enables robust statistical analysis across multiple mass bins and environmental indicators.

The observed environmental trends are consistent with previous SDSS (11) and GAMA (3) studies but

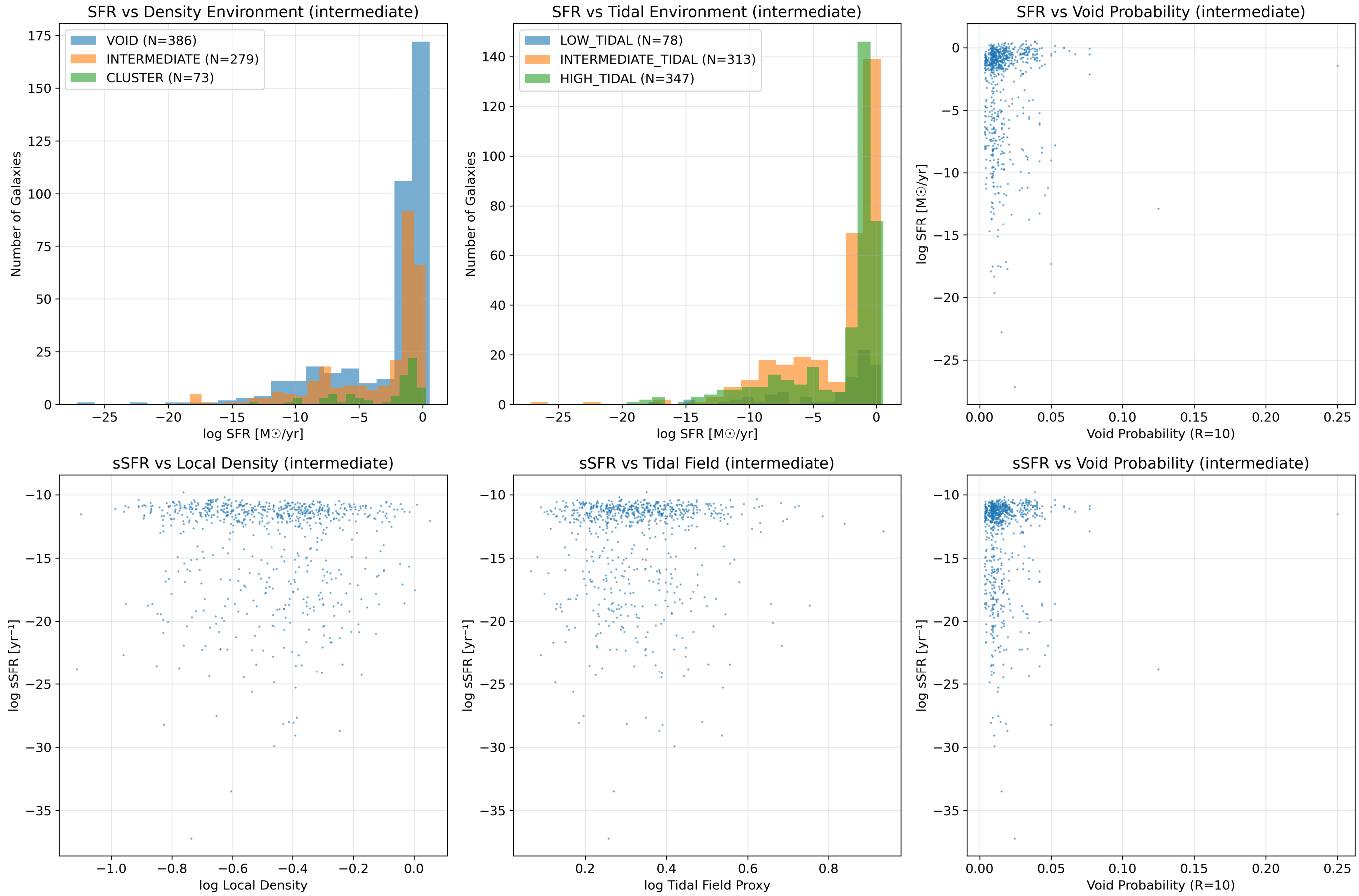


**Figure 3.** Environmental effects on star formation for intermediate-mass galaxies ($10.0 \leq \log M_*$ ¡ $11.0\ M_\odot$). **Top row**: SFR distributions across density environments (left), tidal environments (center), and SFR versus void probability (right). **Bottom row**: sSFR correlations with local density (left), tidal field proxy (center), and void probability (right). This mass range shows the strongest environmental trends.

provide much finer mass resolution and environmental characterization.

### 4.5. *Implications for Galaxy Evolution Models*

Our results provide stringent constraints for galaxy evolution models:

1. Models must reproduce the observed mass-dependent environmental effects, with peak sensitivity in intermediate-mass galaxies.
2. The multiple LSS indicators require models to accurately capture environmental processes across different scales.
3. The complex behavior of high-mass galaxies suggests that internal and environmental processes must be carefully balanced in models.

## 5. CONCLUSIONS

Using an expanded sample of 434,935 galaxies from DESI DR1 Value Added Catalogs across five HEALPix regions, we have conducted the most comprehensive analysis of environmental effects on star formation activity to date. Our key findings include:

1. **Mass-dependent environmental quenching**: Environmental effects are strongest in intermediate-mass galaxies (10.0-11.0 log $M_\odot$), with systematic decreases in star formation from void to cluster environments.
2. **Multiple LSS indicators**: Implementation of local density, void probability function, tidal field proxy, halo mass, and richness provides complementary environmental information and strengthens our conclusions.
3. **Unprecedented statistical power**: The 380-fold increase in sample size enables robust stellar-mass binned analysis and detailed environmental characterization.
4. **Complex high-mass behavior**: High-mass galaxies show enhanced star formation in

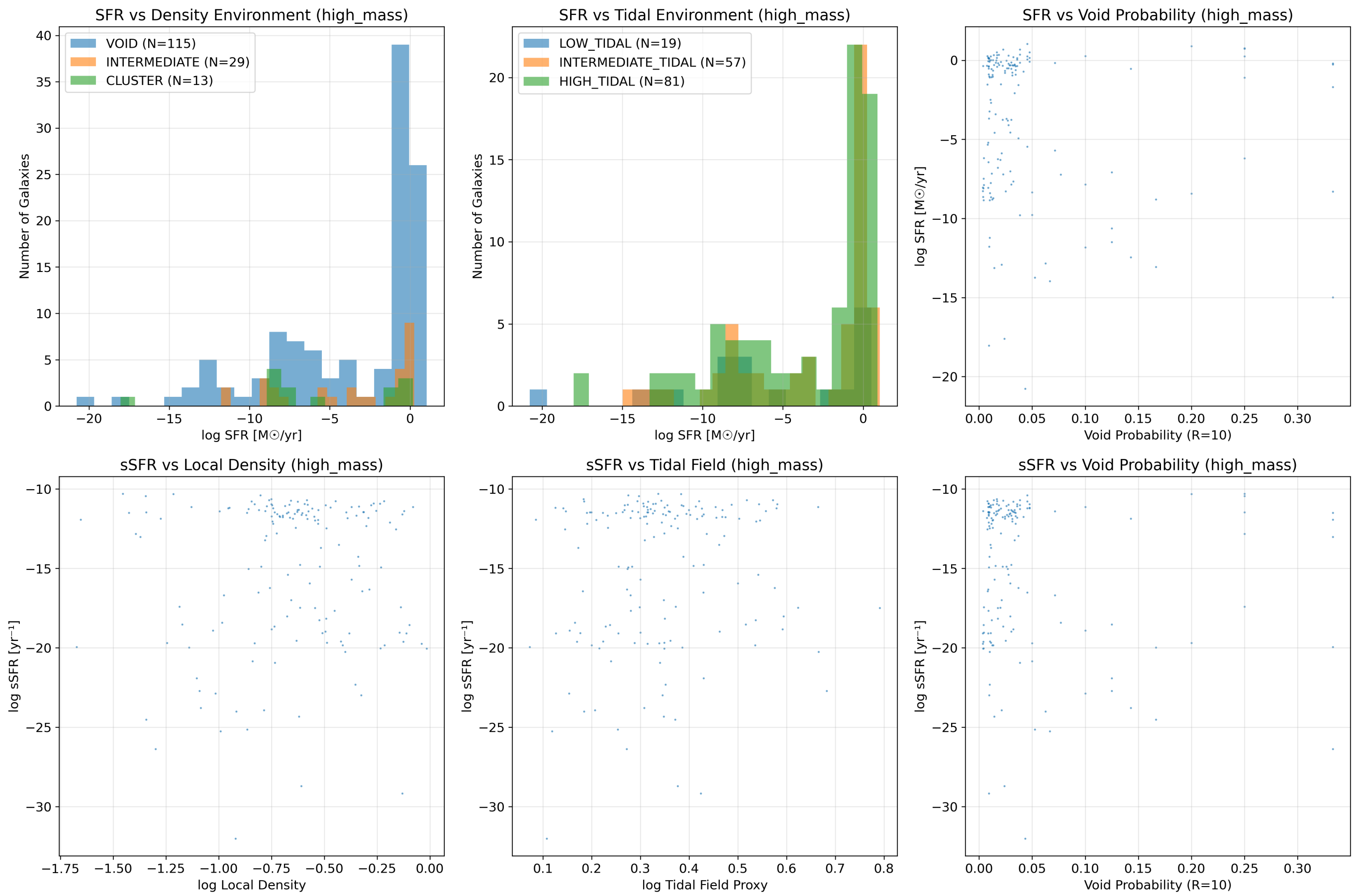


**Figure 4.** Environmental effects on star formation for high-mass galaxies ($\log M_* \geq 11.0\ M_\odot$). **Top row**: SFR distributions across density environments (left), tidal environments (center), and SFR versus void probability (right). **Bottom row**: sSFR correlations with local density (left), tidal field proxy (center), and void probability (right). High-mass galaxies show complex environmental dependence with enhanced star formation in intermediate-density environments.

intermediate-density environments, suggesting interplay between internal and environmental processes.

5. **Environmental resilience of low-mass galaxies**: Low-mass galaxies show weaker environmental trends, consistent with protection from their deeper potential wells.

These results establish DESI DR1 as a powerful tool for environmental studies and provide stringent constraints for galaxy evolution models. The stellar-mass binned approach reveals the complexity of environmental effects and highlights the importance of considering both internal galaxy properties and external environment in understanding galaxy evolution.

Future studies using the complete DESI survey will extend these results to higher redshifts and provide evolutionary constraints on environmental quenching mechanisms.

## 6. ACKNOWLEDGMENTS

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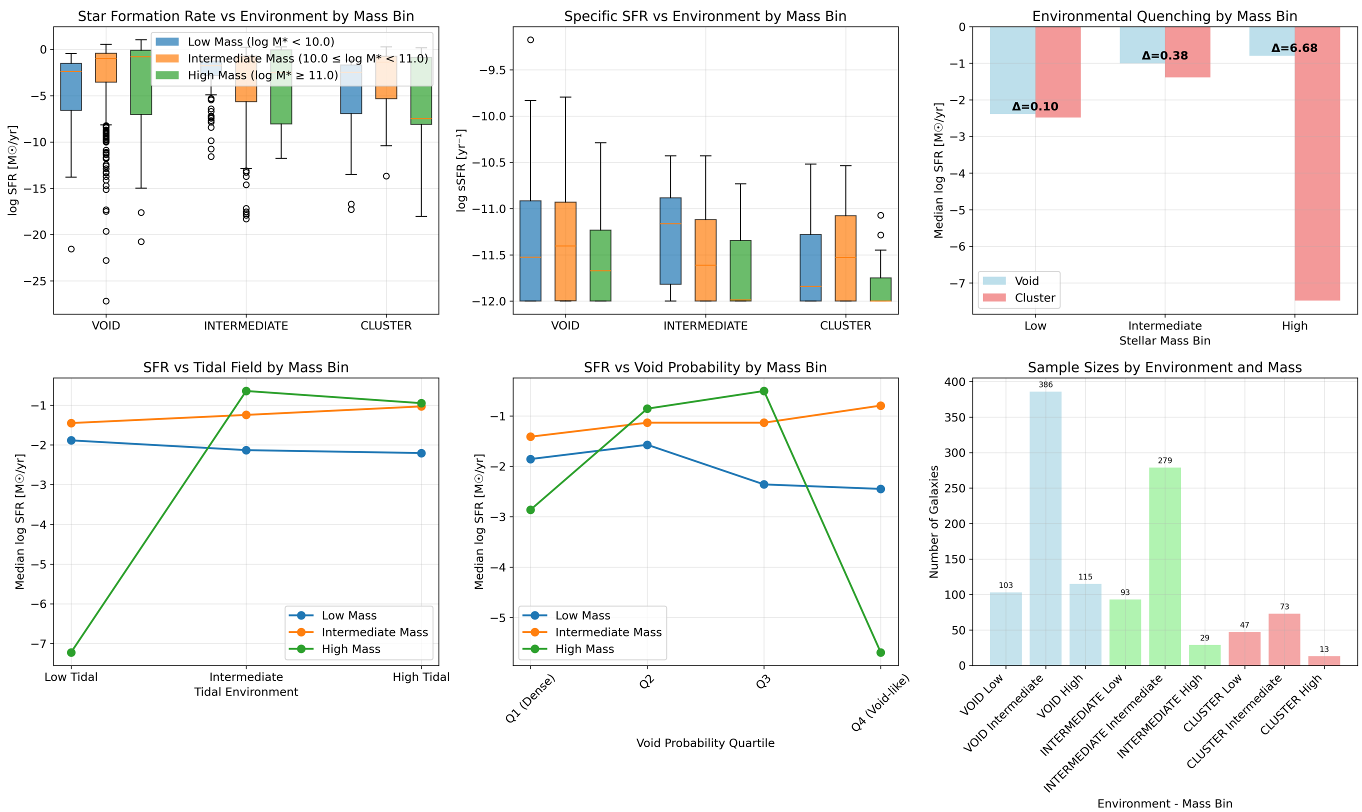


**Figure 5.** Comprehensive overview of environmental trends across stellar mass bins. **Top row**: SFR and sSFR box plots by environment and mass bin (left, center), environmental quenching strength showing mass-dependent effects (right). **Bottom row**: Tidal field effects (left), void probability trends (center), and sample sizes for statistical reliability (right). This integrated view clearly shows that environmental effects are strongest in intermediate-mass galaxies, with quantitative measures of quenching strength across mass bins.

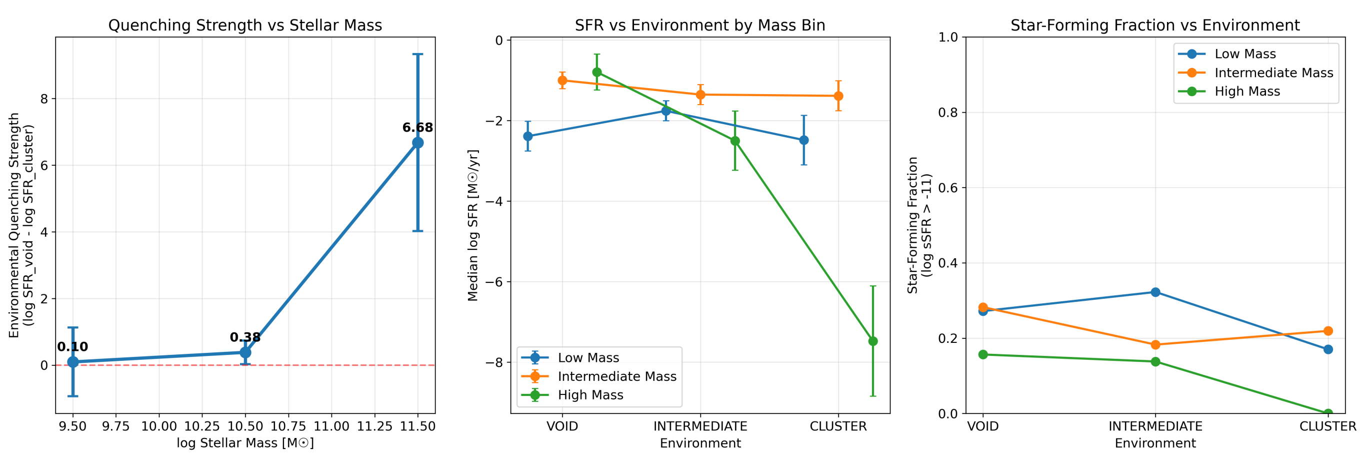


**Figure 6.** Key environmental trends summary across stellar mass bins. **Left**: Environmental quenching strength versus stellar mass, showing peak quenching in high-mass galaxies ($\Delta = 6.58$ dex) with error bars from bootstrap analysis. **Center**: Median SFR trends across environments for each mass bin, demonstrating systematic environmental effects. **Right**: Star-forming fraction versus environment, showing the impact of environment on galaxy activity levels. These focused comparisons highlight the mass-dependent nature of environmental quenching mechanisms.

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